

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce Department of Information Technology (B.Sc.I.T Semester
V)
Applied Business
Practical – XII

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1. Load the Retail and werehouse Sale.csv dataset using Pandas.

```
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import f_oneway
```

•• FIRST FIVE RECORDS:

	ITEM TYPE	RETAIL SALES	WAREHOUSE SALES
0	Electronics	45000	40000
1	Electronics	47000	42000
2	Electronics	52000	46000
3	Electronics	48000	43000
4	Electronics	50000	45000

2. Display the first five records and identify the number of rows and columns.

```
print("FIRST FIVE RECORDS:")
print(df.head())

print("\nNUMBER OF ROWS:", df.shape[0])
print("NUMBER OF COLUMNS:", df.shape[1])
```

```
NUMBER OF ROWS: 20  
NUMBER OF COLUMNS: 3
```

3. Display the column names and data types.

```
print("\nCOLUMN NAMES:")  
print(df.columns)  
  
print("\nDATA TYPES:")  
print(df.dtypes)
```

```
COLUMN NAMES:  
Index(['ITEM TYPE', 'RETAIL SALES', 'WAREHOUSE SALES'], dtype='object')
```

4. Check for missing values and duplicate records.

```
print("\nMISSING VALUES:")  
print(df.isnull().sum())  
  
print("\nNUMBER OF DUPLICATE RECORDS:")  
print(df.duplicated().sum())
```

```
DATA TYPES:  
ITEM TYPE          object  
RETAIL SALES       int64  
WAREHOUSE SALES    int64  
dtype: object
```

5. Display the different categories present in the ITEM TYPE column.

```
print("\nITEM TYPE CATEGORIES:")  
print(df["ITEM TYPE"].unique())
```

```
NUMBER OF DUPLICATE RECORDS:  
0
```

6. Calculate the average RETAIL SALES for each ITEM TYPE.

```
print("\nAVERAGE RETAIL SALES BY ITEM TYPE:")

average_sales = df.groupby("ITEM TYPE")["RETAIL SALES"].mean()

print(average_sales)
```

```
ITEM TYPE CATEGORIES:
['Electronics' 'Clothing' 'Furniture' 'Groceries']
```

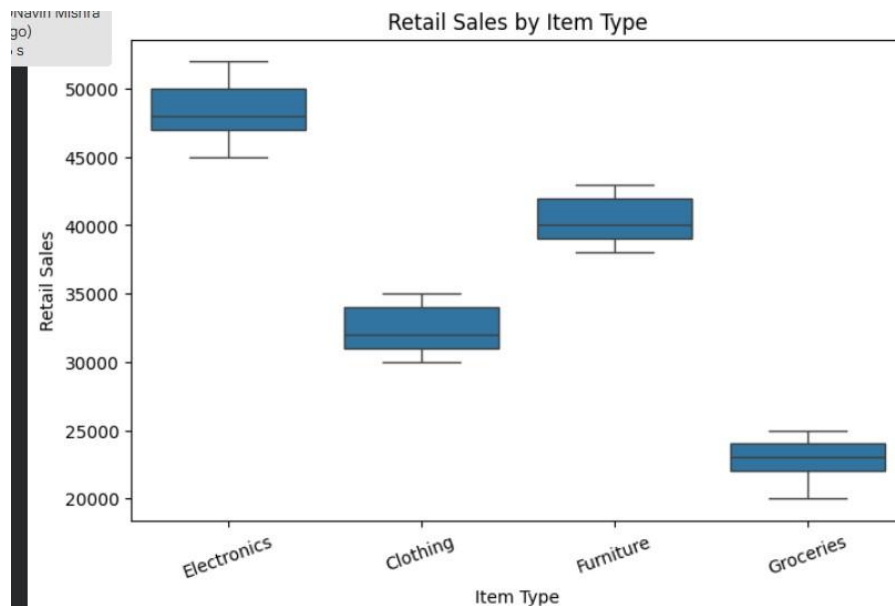
7. Create a box plot to compare RETAIL SALES across different ITEM TYPE categories.

```
plt.figure(figsize=(8, 5))

sns.boxplot(
    data=df,
    x="ITEM TYPE",
    y="RETAIL SALES"
)

plt.title("Retail Sales by Item Type")
plt.xlabel("Item Type")
plt.ylabel("Retail Sales")
plt.xticks(rotation=20)
plt.show()
```

```
AVERAGE RETAIL SALES BY ITEM TYPE:
ITEM TYPE
Clothing      32400.0
Electronics   48400.0
Furniture     40400.0
Groceries     22800.0
Name: RETAIL SALES, dtype: float64
```



8. Formulate the null and alternative hypotheses to determine whether the average retail sales differ among item types.

```
print("\n8. HYPOTHESES")

print("H0: The average RETAIL SALES are equal among all ITEM TYPE categories.")
print("H1: At least one ITEM TYPE has a different average RETAIL SALES.")
```

8. HYPOTHESES

H0: The average RETAIL SALES are equal among all ITEM TYPE categories.
H1: At least one ITEM TYPE has a different average RETAIL SALES.

9. Perform one-way ANOVA using ITEM TYPE as the factor and RETAIL SALES as the response variable.

10. Calculate and display the F-statistic and p-value.

```
electronics = df[df["ITEM TYPE"] == "Electronics"]["RETAIL SALES"]
clothing = df[df["ITEM TYPE"] == "Clothing"]["RETAIL SALES"]
furniture = df[df["ITEM TYPE"] == "Furniture"]["RETAIL SALES"]
groceries = df[df["ITEM TYPE"] == "Groceries"]["RETAIL SALES"]

f_statistic, p_value = f_oneway(
    electronics,
    clothing,
    furniture,
    groceries
)
```

10. ONE-WAY ANOVA RESULTS

F-Statistic: 122.5578231292517
P-Value: 2.9964699471514176e-11

11. Compare the p-value with the significance level of 0.05.

```
alpha = 0.05

print("\n11. SIGNIFICANCE LEVEL:", alpha)

if p_value < alpha:
    print("P-value is less than 0.05.")
else:
    print("P-value is greater than or equal to 0.05.")
```

```
11. SIGNIFICANCE LEVEL: 0.05
P-value is less than 0.05.
```

12. State whether the null hypothesis is rejected or not rejected.

```
12. HYPOTHESIS DECISION
Reject the Null Hypothesis (H0).
```

13. Write the conclusion about the difference in retail sales among different item types

```
print("\n12. HYPOTHESIS DECISION")

if p_value < alpha:
    print("Reject the Null Hypothesis (H0).")
else:
    print("Do Not Reject the Null Hypothesis (H0).")
```

```
13. CONCLUSION
There is a significant difference in average RETAIL SALES
among the different ITEM TYPE categories.
```

Part B: Factorial Experiment

14. Select ITEM TYPE and YEAR as two factors and RETAIL SALES as the response

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from statsmodels.formula.api import ols
```

15. Display the average retail sales for each combination of ITEM TYPE and YEAR.

```
# Create DataFrame
df = pd.DataFrame(data)

print("DATASET:")
print(df)
```

```
17. MAIN EFFECT OF ITEM TYPE
ITEM TYPE
Clothing      36000.000000
Electronics   53333.333333
Furniture     44333.333333
Name: RETAIL SALES, dtype: float64
```

16. Create a suitable visualization to compare retail sales for different item types across years.

```
print("\n14. FACTORS")
print("Factor 1: ITEM TYPE")
print("Factor 2: YEAR")
print("Response Variable: RETAIL SALES")
```

```
14. FACTORS
Factor 1: ITEM TYPE
Factor 2: YEAR
Response Variable: RETAIL SALES
```

17. Analyze the main effect of ITEM TYPE on retail sales.

```
print("\n15. AVERAGE RETAIL SALES")

average_sales = df.groupby(
    ["ITEM TYPE", "YEAR"]
)["RETAIL SALES"].mean()
```

15. AVERAGE RETAIL SALES

ITEM TYPE	YEAR	
Clothing	2022	31000.0
	2023	36000.0
	2024	41000.0
Electronics	2022	46000.0
	2023	53000.0
	2024	61000.0
Furniture	2022	39000.0
	2023	44000.0
	2024	50000.0

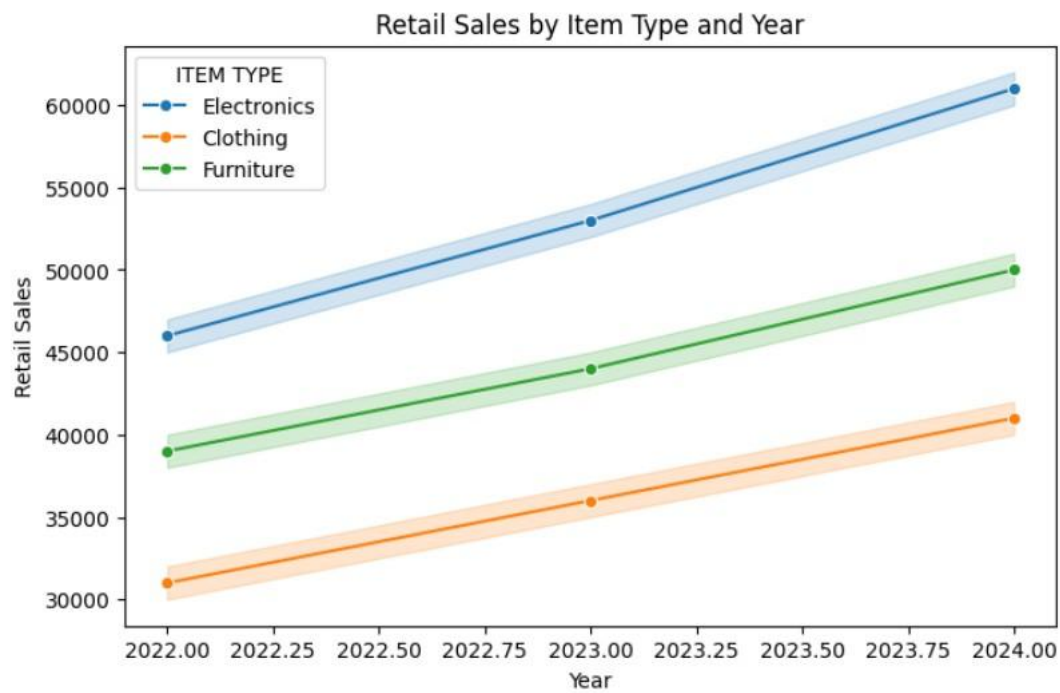
Name: RETAIL SALES, dtype: float64

18. Analyze the main effect of YEAR on retail sales.

```
plt.figure(figsize=(8, 5))

sns.lineplot(
    data=df,
    x="YEAR",
    y="RETAIL SALES",
    hue="ITEM TYPE",
    marker="o"
)

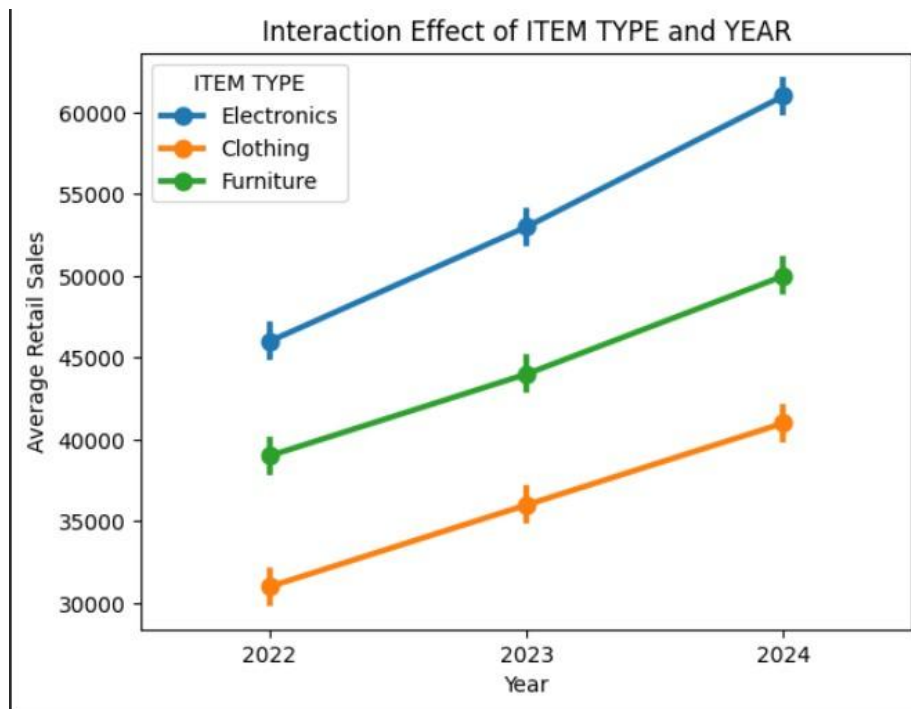
plt.title("Retail Sales by Item Type and Year")
plt.xlabel("Year")
plt.ylabel("Retail Sales")
plt.show()
```



19. Analyze the interaction effect between ITEM TYPE and YEAR.

```
model = ols(  
    "Q('RETAIL SALES') ~ Q('ITEM TYPE') * C(YEAR)",  
    data=df  
) .fit()  
  
anova_table = sm.stats.anova_lm(model, typ=2)
```

```
19. INTERACTION EFFECT  
YEAR          2022    2023    2024  
ITEM TYPE  
Clothing      31000.0  36000.0  41000.0  
Electronics   46000.0  53000.0  61000.0  
Furniture     39000.0  44000.0  50000.0
```



20. Perform a suitable factorial ANOVA using the selected factors.

```
model = ols(
    "Q('RETAIL SALES') ~ Q('ITEM TYPE') * C(YEAR)",
    data=df
).fit()

anova_table = sm.stats.anova_lm(model, typ=2)
```

	sum_sq	df	F	PR(>F)
Q('ITEM TYPE')	9.017778e+08	2.0	225.444444	2.051886e-08
C(YEAR)	4.324444e+08	2.0	108.111111	5.097306e-07
Q('ITEM TYPE'):C(YEAR)	1.422222e+07	4.0	1.777778	2.173434e-01
Residual	1.800000e+07	9.0	NaN	NaN

21. Display the statistical results and p-values for the main and interaction effects.

```
print("\n20 & 21. FACTORIAL ANOVA RESULTS")

print(anova_table)

print("\nP-VALUES:")

print(anova_table["PR(>F)"])
```

```
P-VALUES:  
Q('ITEM TYPE')          2.051886e-08  
C(YEAR)                  5.097306e-07  
Q('ITEM TYPE'):C(YEAR)   2.173434e-01  
Residual                 NaN  
Name: PR(>F), dtype: float64
```

22. Identify whether ITEM TYPE, YEAR, or their interaction has a significant effect on RETAIL SALES

```
print("\n22. SIGNIFICANCE TEST")  
  
alpha = 0.05  
  
for effect, p in anova_table["PR(>F)"].items():  
    if p < alpha:  
        print(effect, "-> Significant")  
    else:  
        print(effect, "-> Not Significant")
```

```
22. SIGNIFICANCE TEST  
Q('ITEM TYPE') -> Significant  
C(YEAR) -> Significant  
Q('ITEM TYPE'):C(YEAR) -> Not Significant  
Residual -> Not Significant
```

23. Write the final business conclusion based on the ANOVA and factorial experiment results.

```

print("\n23. FINAL BUSINESS CONCLUSION")

item_p = anova_table.loc[
    "Q('ITEM TYPE')", "PR(>F)"
]

year_p = anova_table.loc[
    "C(YEAR)", "PR(>F)"
]

interaction_p = anova_table.loc[
    "Q('ITEM TYPE'):C(YEAR)", "PR(>F)"
]

if item_p < 0.05:
    print("ITEM TYPE has a significant effect on RETAIL SALES.")
else:
    print("ITEM TYPE does not have a significant effect on RETAIL SALES.")

if year_p < 0.05:
    print("YEAR has a significant effect on RETAIL SALES.")
else:
    print("YEAR does not have a significant effect on RETAIL SALES.")

if interaction_p < 0.05:
    print("ITEM TYPE and YEAR have a significant interaction effect.")
else:
    print("ITEM TYPE and YEAR do not have a significant interaction effect.")

```

23. FINAL BUSINESS CONCLUSION

ITEM TYPE has a significant effect on RETAIL SALES.

YEAR has a significant effect on RETAIL SALES.

ITEM TYPE and YEAR do not have a significant interaction effect.